

OBSERVE · INVESTIGATE · VERIFY

ANVEŞHAI

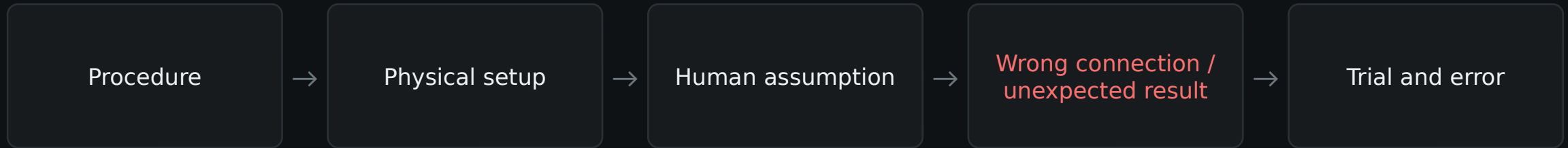
On-Device Intelligence for Physical Experiment Verification

“AI that verifies what you built, not just what you typed.”

Snapdragon AI Lab Build & Present Challenge

THE PROBLEM

Procedures tell you what to do. Nothing checks what you built.

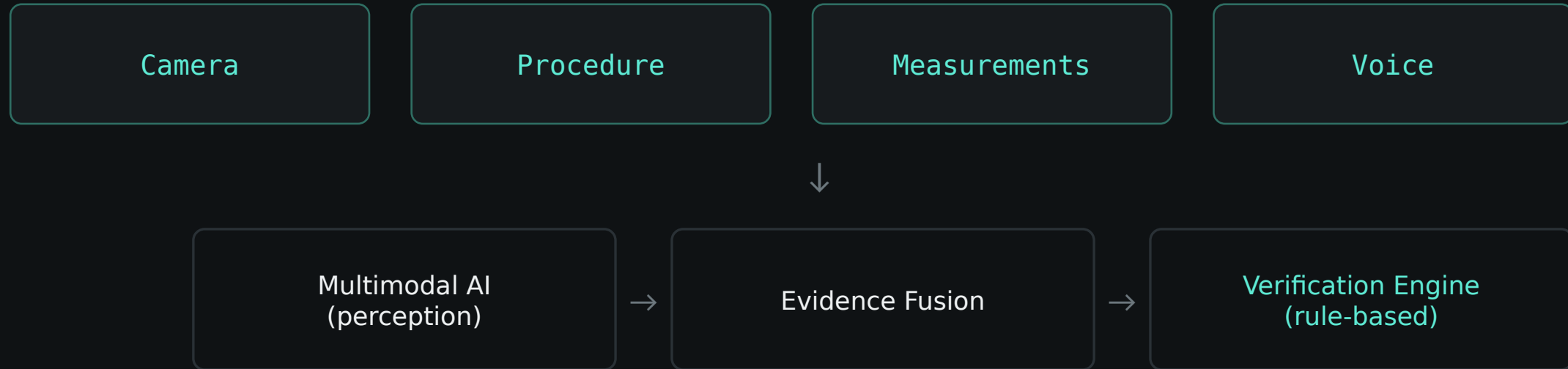


A wrong pin, a reversed component, an out-of-range reading — caught today only by a TA walking the room, or by hours of undirected debugging.

Digital instructions know what should happen.

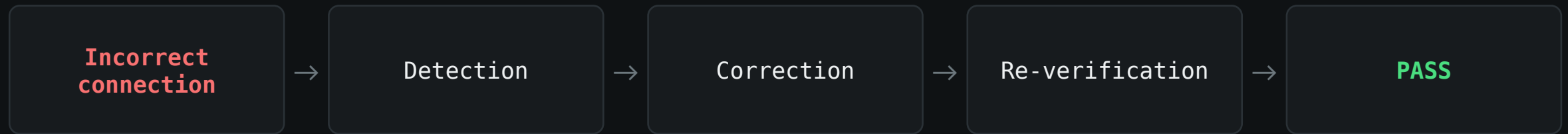
ANVEṢHA understands what is actually happening.

Evidence fusion, not model opinion



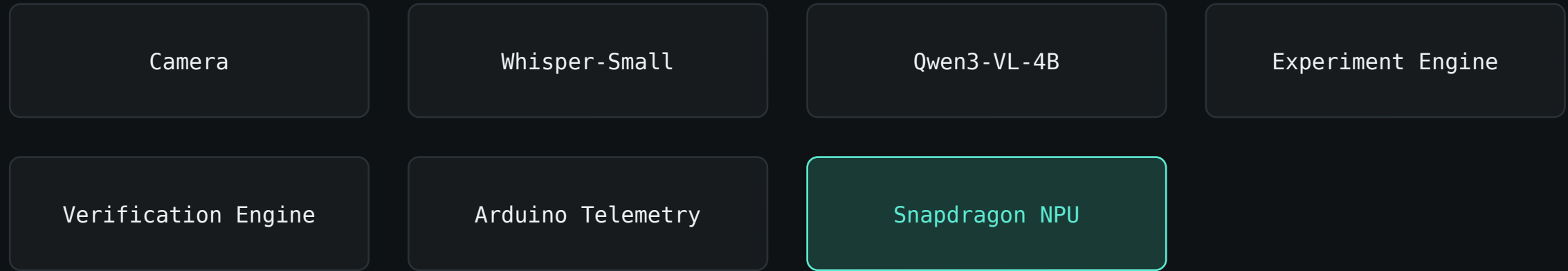
Perception answers “what do I see?” — the verification engine answers “does that satisfy the experiment?”

Wrong pin to verified — live



LDR sensor OUT wired to A1 instead of A0. ANVEŞHA flags the deviation, names the exact pin mismatch, and confirms PASS the moment it's corrected.

Clean separation: perception vs. verification



Local AI is the product, not a feature flag

Privacy

Camera, audio, and experiment data never leave the device.

Low latency

A tight capture → verify → correct loop needs fast, local inference.

Offline inference

Works in labs and classrooms with no reliable network.

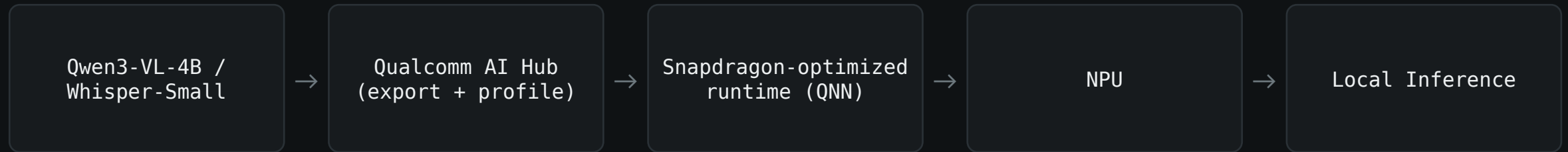
Efficient continuous inference

NPU acceleration for repeated frame-by-frame checks.

Combine multiple AI workloads

Vision + speech running together without cloud round-trips.

Model → AI Hub → NPU → Local Inference



This repository implements the full backend abstraction (device detection, runtime selection, honest unavailability) and a verified, sourced deployment guide ([docs/qualcomm_deployment.md](#)). NPU execution itself requires physical Snapdragon hardware not available during development — no APIs or benchmarks are fabricated.

Measured results vs. AI Hub reference metrics

MEASURED (this repo)

CPU fallback vision backend

Real OpenCV frame-quality gating

- Load time
- First-inference latency
- p50 / p95 vision latency
- Verification engine latency
- Peak memory

See `benchmarks/results/benchmark_report.md`
for exact current numbers.

AI HUB REFERENCE (not ours)

No Snapdragon NPU is available in this development environment. Qualcomm's own on-device profiling (run automatically during model export via `qai_hub_models`) is the correct source for NPU figures — kept in a clearly separate report section, never merged with our measurements.

Where this goes next

APPLICATIONS

- Engineering education
- University laboratories
- Maker spaces
- Technical training
- Equipment setup verification
- Remote / asynchronous learning

FUTURE WORK

- More experiments
- Better component recognition (Qwen3-VL on-device)
- Automatic report generation
- Additional sensors
- Advanced multimodal reasoning